Subject :

T.P.H Date :
Time :

① Show that 2 is a primitive root modulo 11.

Ans:

In modular arithmetic, a number g is called a primitive root modulo n if every number a coprime to n is congruent to a power of g modulo n.

Mathematically,

$\gcd(a, n) = 1$ there exists an integer k such that $g^k \equiv a \pmod{n}$.

Let g be a primitive root modulo n, then we can show that the smallest number k for which $g^k \equiv 1 \pmod{n}$ is equal to $\phi(n)$.

$$\mathrm{ord}_n(g) = \phi(n) \qquad n \in \mathbb{N}$$

where,

$\mathrm{ord}_n(g)$ = smallest positive integer k that $g^k \equiv 1 \pmod{n}$.

$\phi(n)$ = Euler's totient function.

Now, for 2 to be a primitive root modulo 11 we need to demonstate that the order of 2 modulo 11 is equal to $\phi(11)$.

since 11 is a prime number,

the Euler's totient function value is:

$$\phi(11) = 11 - 1 = 10.$$

CamScanner

Subject :

T.P.H Date : Time :

For 2 to be a primitive root, the smallest positive integer k such that $2^k \equiv 1 \pmod{11}$ must be 10. and the power of 2 ~~runs~~ has to run through all numbers from 1 to 10. $\{1, 2, \ldots, 10\}$.

Verification By listing powers

(powers of 2 modulo 11)

k	$2^k \pmod{11}$
1	2
2	4
3	8
4	5
5	10
6	9
7	7
8	3
9	6
10	1

so we obtain $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Exactly all non-zero residues modulo 11.

since $\text{ord}_{11}(2) = 10 = \phi(11)$

we can conclude 2 is a primitive ~~m~~ root modulo 11. and its power run through every non-zero residue modulo 11.

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Subject :

T.P.H Date :
Time :

② How many incongruent primitive roots does 14 have?

Ans:
A modulus n has primitive roots if and only if n is 2, 4, p^k or $2p^k$ (where p is an odd prime).

Since 7 is an odd prime, 14 fits the form $2p^1$.
Therefore primitive roots ~~no~~ modulo 14 do exist.

The number of incongruential ~~primitive~~ primitive roots modulo n is given by:

$\phi(\phi(n))$. where ϕ is the Euler's ~~totient~~ function.

Since ϕ multiplicative for coprime factors:

$$\phi(14) = \phi(2) \times \phi(7)$$

$$\phi(14) = (2-1)(7-1)$$

$$= 6.$$

This tells us there are 6 integers coprime to 14:
$\{1, 3, 5, 9, 11, 13\}$.

Calculating $\phi(\phi(n))$

$$\phi(6) = \phi(2) \times \phi(3)$$

$$= (2-1)(3-1)$$

$$= 2.$$

There are 2 incongruent primitive roots modulo ~~n~~ 14.

Subject :

T.P.H Date :
Time :

These primitive roots are 3 and 5. All Elements coprime to 14 will have an order less than 6.

③ Suppose n is a positive integer and a^{-1} is a multiplicative inverse of $a \pmod{n}$.

ⓐ Show $\text{ord}_n a = \text{ord}_n(a^{-1})$.

Ans:

To show two numbers are equal, we can show each is less than or equal to the other, or simply show they satisfy the same definition.

Let, $k = \text{ord}_n(a)$ and $m = \text{ord}_n(a^{-1})$.

~~Case 1~~

$(m \leq k)$:

~~since~~ $\text{ord}_n(a) = k$

$\therefore a^k \equiv 1 \pmod{n}$ (where k is the smallest positive integer with this property)

Now taking inverses on both sides:

$$(a^k)^{-1} \equiv (1)^{-1} \pmod{n}$$

$$\Rightarrow (a^{-1})^k \equiv 1 \pmod{n}$$

since $(a^{-1})^k = 1$, the order of (a^{-1}) ~~dev~~ divides k.

Subject :

T.P.H Date : Time :

so: $\text{ord}_n(a^{-1}) \leq k$.

$m \leq k$.

~~Ie: Case-2~~

$k \leq m$:

since $\text{ord}_n(a^{-1}) = m$

Then $(a^{-1})^m \equiv 1 \pmod{n}$.

Taking inverses again,

$((a^{-1})^m)^{-1} \equiv 1^{-1} \pmod{n}$.

$\Rightarrow a^m \equiv 1 \pmod{n}$.

So, the order of a divides m

$\therefore \text{ord}_n(a) \leq m$

$\therefore k \leq m$.

We have $m \leq k$ and $k \leq m$ Therefore, $k = m$,

$$\boxed{\text{ord}_n(a) = \text{ord}_n(a^{-1})}$$

Subject :

T.P.H Date :
Time :

(b) If a is primitive root modulo n, must a^{-1} also be a primite root?

Ans:

A primitive root modulo n is defined as an element whose order is exactly $\phi(n)$.

If a is a primitive root modulo n, then:

$$\text{ord}_n(a) = \phi(n).$$

from part (a) we got:

$$\text{ord}_n(a) = \text{ord}_n(a^{-1}).$$

so, $\text{ord}_n(a^{-1}) = \phi(n)$.

Since the order of a^{-1} is also $\phi(n)$, a^{-1} must be a primitive root modulo n.

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